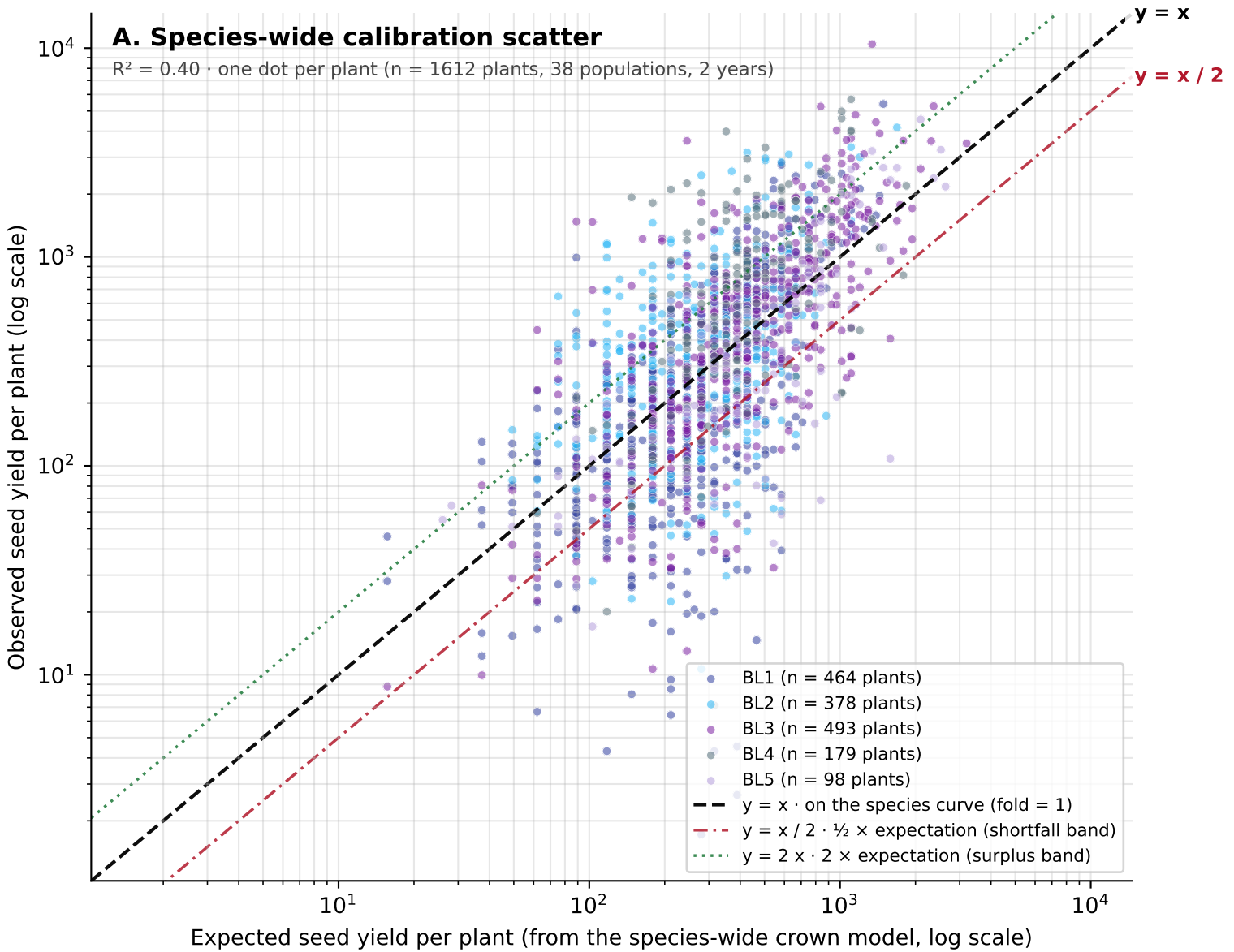
Species-wide plant-size → seed-yield expectation (the yardstick for the per-population seed-yield test)



B. Model build — how the species-wide expectation is set

STAGE 1 · predictor selection (10-fold cross-validation)

Lower CV-RMSE = better out-of-sample fit → wins Stage 2.

predictor	CV-RMSE	±sd	R^2_in	AIC	
crown	0.410	0.031	0.400	1713.6	⋆
crown+height	0.411	0.031	0.400	1715.6	
area	0.421	0.027	0.370	1794.1	
height	0.454	0.026	0.269	2033.3	

→ Winner: crown (★)

STAGE 2 · refitted expectation model (all plants pooled)

Fitted value = expected $\log_{10}$(yield) for a plant of that size.
Residual = observed − expected.

Formula : $\log_{10}$(yield) ~ logc
Coefficients : Intercept = +1.195, logc = +1.252
R^2 (in sample): 0.400
Residual SD : 0.411 $\log_{10}$ (≈ × / ÷ 2.6)

Every population in the per-population test is measured against this expectation.